# EXPLORATORY DATA ANALYSIS REPORT

Project 2 — Cafe Sales Dataset | DecodeLabs Batch 2026

Dataset: 300 transactions | Period: Jan–Dec 2023 | Tool: Python (pandas, matplotlib)

## Business Overview

| TOTAL REVENUE | TOTAL TRANSACTIONS | MEAN SPEND | MEDIAN SPEND |
|---|---|---|---|
| **$3,536.50** | **300** | **$11.79** | **$10.50** |
| **TOP ITEM** | **TOP CATEGORY** | **PEAK MONTH** | **TOP PAYMENT** |
| **Sandwich** | **Food** | **10** | **Debit Card** |

## 1. Descriptive Statistics — The Five-Number Summary

*Python tip: df['column'].describe() — gives the logic skeleton of any variable.*

| Metric | Quantity | Price Per Unit ($) | Total Spent ($) |
|---|---|---|---|
| **Count** | 300 | 300 | 300 |
| **Mean** | 2.94 | 4.07 | 11.79 |
| **Std Dev** | 1.34 | 1.29 | 6.45 |
| **Min** | 1 | 2.50 | 2.50 |
| **Q1 (25th)** | 2 | 3.00 | 7.00 |
| **Median (50th)** | 3 | 3.50 | 10.50 |
| **Q3 (75th)** | 4 | 4.50 | 16.00 |
| **Max** | 5 | 7.00 | 35.00 |
| **Skewness** | **0.043 (symmetric)** | **—** | **0.818 (right-skewed)** |

**Insight:** Total_Spent is right-skewed (0.818) — the mean ($11.79) is pulled above the median ($10.50) by large orders. Use the **median as the true center** for business decisions (e.g. pricing, promotions).

## 2. Outlier Detection — IQR Method

| Column | Q1 | Q3 | IQR | Lower Bound | Upper Bound | Outliers | Classification |
|---|---|---|---|---|---|---|---|
| Total_Spent | $7.00 | $16.00 | $9.00 | -$6.50 | $29.50 | 4 | **SIGNAL** |
| Quantity | 2 | 4 | 2 | -1 | 7 | 0 | None |

**Insight:** 4 outlier transactions exceed $29.50. These are classified as **SIGNAL** (not noise) — likely large group orders (e.g. 5× Salad = $35.00). Action: *Investigate, do not remove.*

## 3. Item & Category Performance

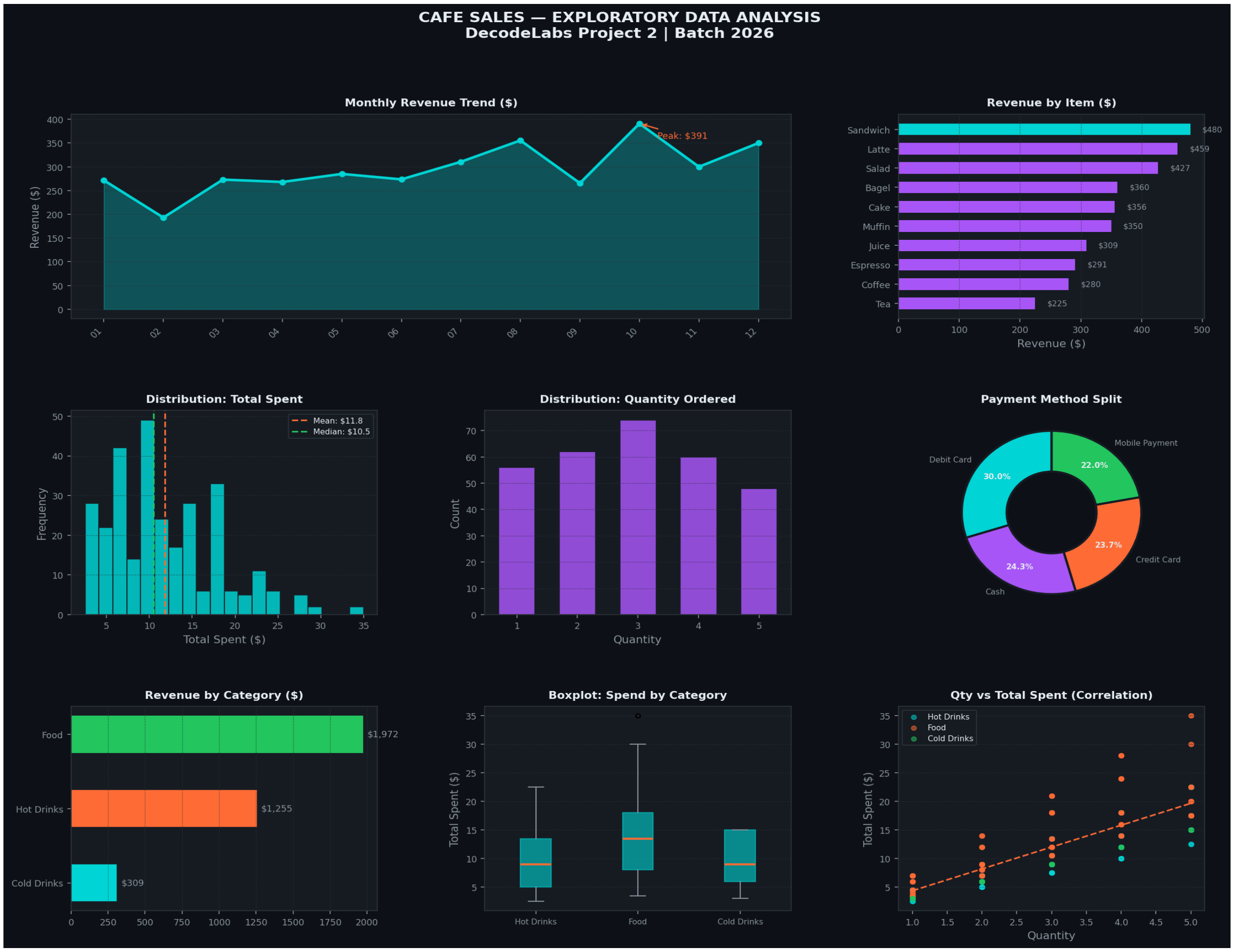
| Item | Revenue ($) | Transactions | Avg Order ($) |
|---|---|---|---|
| **Sandwich** | **$480.00** | **29** | **$16.55** |
| Latte | $459.00 | 35 | $13.11 |
| Salad | $427.00 | 24 | $17.79 |
| Bagel | $360.00 | 28 | $12.86 |
| Cake | $355.50 | 27 | $13.17 |
| Muffin | $350.00 | 33 | $10.61 |
| Juice | $309.00 | 31 | $9.97 |
| Espresso | $291.00 | 35 | $8.31 |
| Coffee | $280.00 | 28 | $10.00 |
| Tea | $225.00 | 30 | $7.50 |

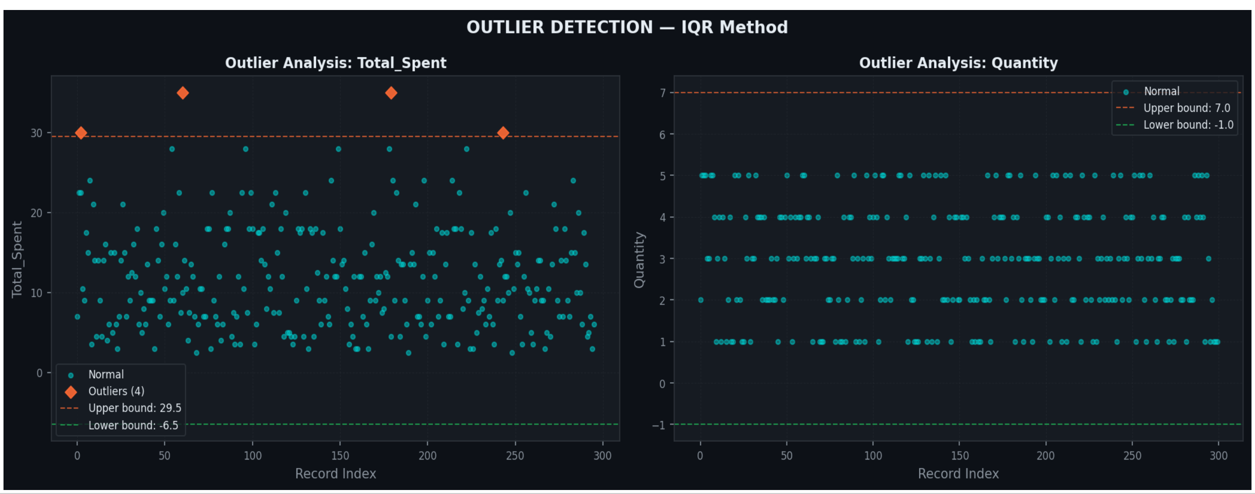
## 4. Correlation Analysis

| Variable Pair | Pearson r | Strength | Interpretation |
|---|---|---|---|
| Quantity ↔ Total_Spent | 0.80 | Strong Positive | More items ordered → higher bill (expected) |
| Price_Per_Unit ↔ Total_Spent | 0.47 | Moderate Positive | Pricier items contribute to higher spend |
| Quantity ↔ Price_Per_Unit | -0.09 | No Correlation | People order same qty regardless of price |

**Golden Rule:** Correlation ≠ Causation. A correlation is a **CLUE**, not a VERDICT. Always check for hidden variables (confounders) before drawing conclusions.

## 5. Visual Evidence




## 6. Executive Summary — The 'So What?' Test

**Problem Statement:** What drives revenue at the cafe and where are the growth opportunities?

**Methodology:** EDA on 300 cleaned transactions using descriptive statistics, IQR outlier detection, skewness analysis, correlation analysis, and monthly trend analysis.

| | |
|---|---|
| **Finding 1 — Food dominates revenue (56%)** | Food category generates $1,972.50 (56% of total revenue) despite representing only 47% of transactions. This signals a higher average order value vs Hot/Cold Drinks. |
| **Finding 2 — Salad has highest avg order ($17.79)** | Salad and Sandwich command the highest average order values. These items are natural upsell targets when bundled with drinks. |
| **Finding 3 — Revenue is right-skewed (skewness = 0.82)** | The mean ($11.79) exceeds the median ($10.50), indicating a few large orders inflate averages. Median is the better central measure for pricing decisions. |
| **Finding 4 — 4 outlier transactions detected (SIGNAL)** | All 4 outliers exceed $29.50 and represent large group orders. These are valuable customers — not data errors. Action: create a 'group order' loyalty program. |
| **Finding 5 — Strong Qty ↔ Revenue correlation (r = 0.80)** | Volume is the primary driver of spend. Promotions that increase quantity per visit (bundles, combos) will directly grow revenue. |
| **Finding 6 — Debit Card leads in avg spend ($12.44)** | Debit Card users spend more per transaction on average. Mobile Payment users have the lowest avg spend ($11.70) — digital promotions could close this gap. |

**Recommendations (Actionable Next Steps):**

1. Launch Sandwich + Coffee bundle deal to increase avg order value across categories
2. Introduce group order loyalty cards targeting the 4 high-spend outlier customer profiles
3. Run Mobile Payment exclusive promotions to close the $0.74 avg-spend gap vs Debit Card
4. Plan inventory peaks for Food items — highest revenue and transaction volume
5. Next step: Build a visualization dashboard (Project 3) to monitor these KPIs in real time

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*Dataset: Cafe Sales | Tool: Python (pandas, matplotlib, scipy) | DecodeLabs Industrial Training Kit | Batch 2026*